

Topics: 12.5 Equations of Lines and Planes (Part II); 12.6 Cylinders and Quadric Surfaces; 13.1 Vector Functions and Space Curves

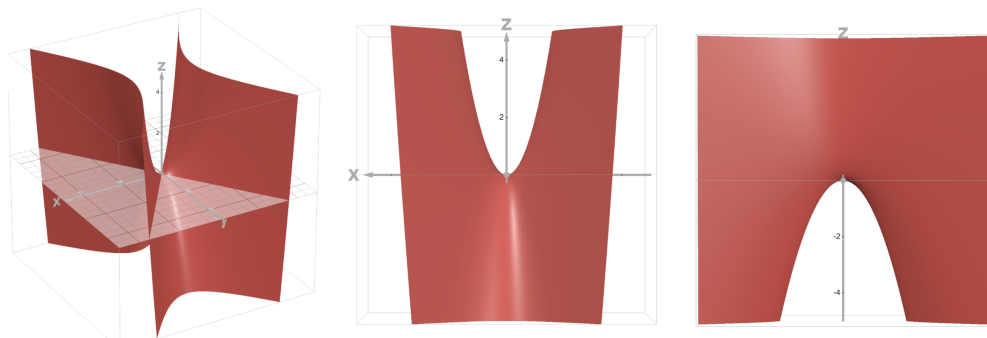
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- Find parametric equations for the line of intersection of the planes

$$2x + 3y + 5z = 7 \quad \text{and} \quad x - y + 2z = 3.$$

- Find an equation of the plane through $P(7, -2, -4)$ that is parallel to the plane $z = 4x - 5y$.
- Find an equation of the plane that passes through the point $P(10, -1, 5)$ and contains the line with symmetric equations $\frac{x}{4} = y + 6 = \frac{z}{5}$.
- Determine whether the line L given by $\mathbf{r}(t) = \langle -2t, 2 + 7t, -1 - 4t \rangle$ intersects the plane given by $4x + 9y - 2z + 8 = 0$.
- A surface is shown below from three different viewpoints.



Which of the following equations represents the surface?

- $x^2 + y^2 + z^2 = 1$
- $z = 1 - x^2 - 2y^2$
- $z^2 = x^2 + y^2$
- $z = 2x^2 - y^2$

- Find the domain of the vector function

$$\mathbf{r}(t) = \left\langle \ln(t+1), \frac{t}{4-t^2}, \sqrt{4-t} \right\rangle.$$

- Evaluate the limit $\lim_{t \rightarrow 1} \mathbf{r}(t)$, where

$$\mathbf{r}(t) = \left\langle \frac{t^2 - 1}{t^2 - 3t + 2}, \frac{t - 1}{\sqrt{t + 3} - 2}, \frac{\sin(t - 1)}{t - 1} \right\rangle.$$

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Equations of a plane. The plane through (x_0, y_0, z_0) with normal vector $\langle a, b, c \rangle$ is given by

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

By collecting terms, we can rewrite this equation in the form

$$ax + by + cz = d$$

where $d = ax_0 + by_0 + cz_0$ is a constant. Two planes are parallel if their normal vectors are parallel, and they are perpendicular if their normal vectors are orthogonal.

Vector functions. A **vector-valued function** (or **vector function**) is a function whose domain is a set of real numbers and whose range is a set of vectors. In three dimensions, a vector-valued function is written as

$$\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k},$$

where f , g , and h are real-valued **component functions**. The domain of $\mathbf{r}(t)$ is the set of all values t for which $f(t)$, $g(t)$, and $h(t)$ are all defined. Limits of vector functions are defined component-wise: For a real number a ,

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \left\langle \lim_{t \rightarrow a} f(t), \lim_{t \rightarrow a} g(t), \lim_{t \rightarrow a} h(t) \right\rangle,$$

provided the three component limits exist.

Suggested Textbook Problems

Section 12.5: 1-72

Section 12.6: 1-49

Section 13.1: 1-58